

Test agents in a Slack-like workspace



Slack workflows



Automatic result checks

- Relay tests whether an agent can complete a task in a working web app.
- 18 tasks cover search, threads, message edits, direct messages, and channel settings.
- Each run records the actions, checks the result, and estimates the cost.
- The goal is to complete tasks with fewer errors, less time, and lower cost.



Task



Private workspace



Agent actions

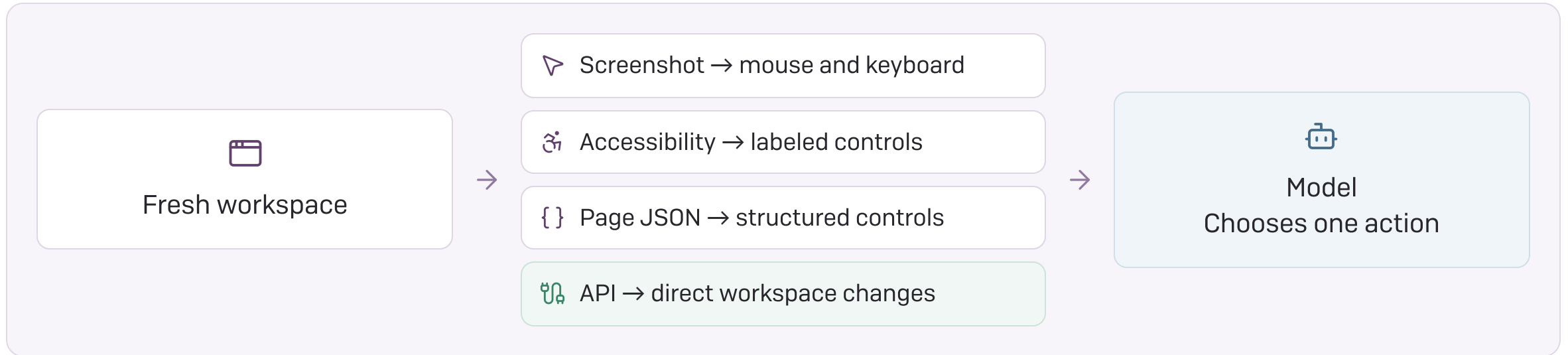


Result checks

The harness runs and records the test

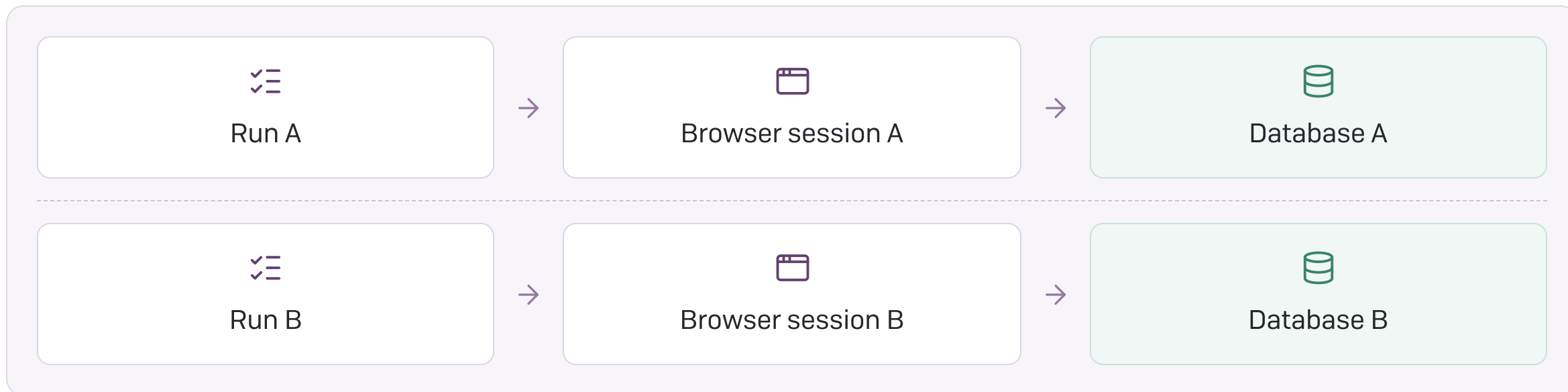
- The harness is the code that controls each test run.
 - It shows the workspace to the model and asks for one action.
 - It checks the action, executes it, and records what changed.
 - It repeats these steps until the model finishes or reaches a limit.
 - It checks the final workspace against the task requirements.
-
- React builds the UI. SQLite stores workspace data. Playwright controls the browser.
 - [Related projects and references](#). I reviewed their code but did not run a comparison.

Four ways to use the workspace



- The model receives the task and workspace information, but not the expected answer.
- API runs test tool use without browser interaction. They are reported separately.
- The 306-run comparison uses accessibility controls for every model.

Each run has its own workspace



- Each run starts with a new browser session and a separate SQLite database.
- Changes in one workspace do not change another workspace.
- Version checks reject outdated edits. Request IDs prevent duplicate writes.
- This separates workspace data. It does not safely contain malicious code.

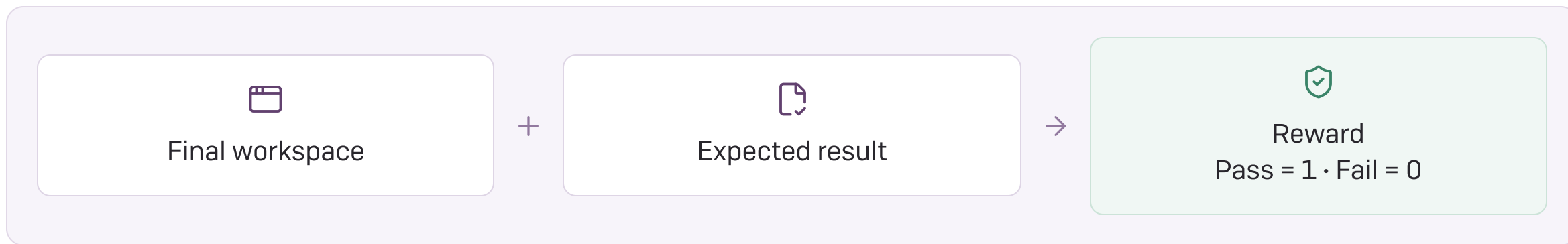
Run Relay with different models

 17 models 18 tasks 1 run per task and model

- Use Node 24. Follow the README to install, build, and start Relay.
 - Add a Router API key, then choose an available model and a task.
 - The comparison uses the same task data, controls, and result checks for every model.
 - Each run allows up to 40 actions, 180 seconds, and a \$5 estimated cost.
 - The manual workspace and saved replays do not require an API key.
-
- [Exact settings and run history.](#)

Define the task and check the result

- The suite has six simple tasks and 12 tasks that require several changes.
- Example: find an approval, update the channel description, pin the message, save the reply, and acknowledge it.
- The checks verify the text, author, destination, and required changes.
- A run fails if it leaves a duplicate message or changes unrelated data.



- These checks grade the final state. They do not penalize a wrong action that the agent fully undoes.

Test the app and the result checks

- 174 backend tests and 91 browser tests check the software.
- 84 valid results and 2,583 deliberately invalid results test the checks for the 12 longer tasks.
- Scripted tests verify the UI. They do not show what a model can do.
- 306 recorded model runs show how the agents performed.



Software tests
Does the app work?



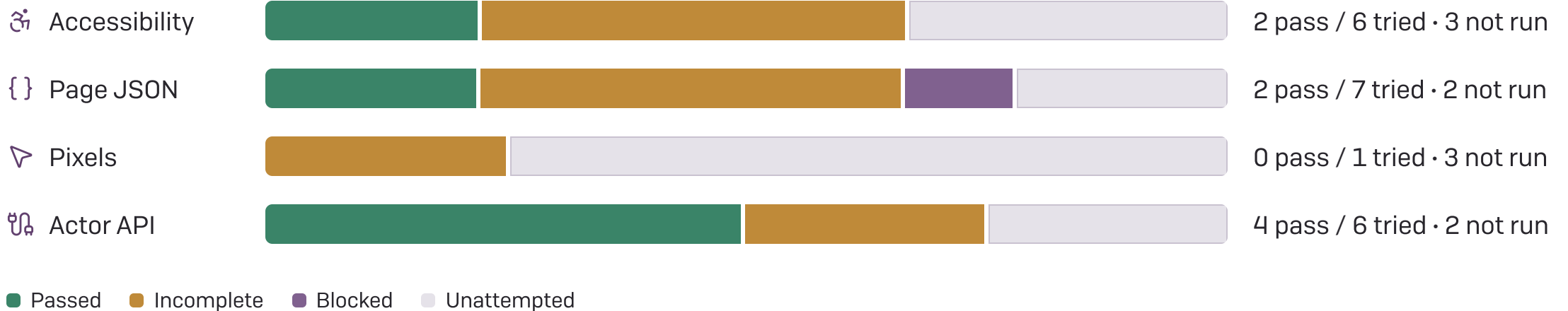
Result tests
Are incorrect results rejected?



Model runs
Can the agent do the task?

Results from the earlier interface tests

- 20 attempted: 8 passed, 11 incomplete, and 1 blocked. 10 planned runs did not start.
- These tests used different tasks and models, so they cannot establish which interface is better.
- These results are separate from the 306-run model comparison.



- 126 requests cost an estimated \$0.149874, including a \$0.001490 reservation for missing usage. This is not an invoice.
- [All interface runs and recorded actions.](#)

Results for 17 models

1 per task/model · Collection complete

306 / 306 attempted · 152 passed · 61 incomplete · 93 blocked

\$189.3110

Usage estimate

\$3.7794

Unresolved · 35 calls

\$193.0904

Total allowance · 306 trials

Model ↕	Trials ↕	Passed ↕	Incomplete ↕	Blocked ↕	Median time ↕	Usage estimate ↕	Unresolved ↕	Total allowance ↕
 GPT-6 Astra	18 / 18	14 / 18	1	3	27.1s	\$35.9623	\$0.0000	\$35.9623
 GPT-6.1 Sol	18 / 18	14 / 18	4	0	28.5s	\$8.4352	\$0.0000	\$8.4352
 GPT-6 Luna	18 / 18	9 / 18	8	1	39.4s	\$0.5394	\$0.0092	\$0.5486
 GPT-OSS 120B	18 / 18	5 / 18	2	11	180.1s	\$0.6682	\$0.1358	\$0.8040
 Claude Fable 5.1	18 / 18	13 / 18	1	4	62.8s	\$48.4714	\$0.0000	\$48.4714
 Claude Opus 5.5	18 / 18	13 / 18	4	1	38.3s	\$28.6987	\$0.3741	\$29.0728
 Claude Sonnet 5.5	18 / 18	7 / 18	4	7	39.9s	\$13.9323	\$0.3304	\$14.2627
 Claude Haiku 4.5	18 / 18	6 / 18	3	9	28.3s	\$3.7921	\$0.0000	\$3.7921
 Grok 4.7	18 / 18	10 / 18	6	2	34.3s	\$11.9101	\$0.3809	\$12.2910
 Qwen 3.8 Max	18 / 18	7 / 18	1	10	125.1s	\$6.9999	\$1.0639	\$8.0638
 DeepSeek V4 Pro	18 / 18	8 / 18	1	9	106.8s	\$4.6816	\$0.2547	\$4.9363
 DeepSeek V4.1 Flash	18 / 18	7 / 18	3	8	65.9s	\$1.3825	\$0.0541	\$1.4365
 GLM 5.3	18 / 18	7 / 18	0	11	125.3s	\$3.6307	\$0.3453	\$3.9760
 GLM 5.3 Flash	18 / 18	8 / 18	0	10	46.2s	\$0.4444	\$0.0000	\$0.4444
 Kimi K3	18 / 18	11 / 18	3	4	50.0s	\$14.0051	\$0.7958	\$14.8009
 MiniMax M3	18 / 18	5 / 18	10	3	25.9s	\$1.4698	\$0.0353	\$1.5052
 Nemotron 3 Ultra	18 / 18	8 / 18	10	0	50.1s	\$4.2874	\$0.0000	\$4.2874

Passed

Incomplete

Blocked

Not run

- 18 public tasks. 1 run per task and model. All use accessibility controls. Median time includes blocked runs.
- Total allowance adds the usage estimate to money reserved for missing usage. These are estimates, not invoices. Including earlier tests: \$194.1058 of the \$300 budget. [Full costs CSV](#) · [JSON](#)
- 105 earlier runs are included. No retries. One run per task cannot establish a reliable model ranking. Two tasks have unclear instructions. [Read the failure analysis](#).
- [Test plan](#) · [Recorded runs](#) · [Download all trials \(CSV\)](#) · [Earlier 36-trial pilot: 20 passed](#)

My main mistake: I did not validate the task instructions

- I tested whether the UI and grading code worked, but did not independently validate every task instruction.
 - Design handoff quoted “DESIGN” without requiring replacement. Six completed runs copied it, but the grader expected “Willow”.
 - Release sync gave an unclear channel location for one source message.
 - These tasks do not give a clear measure of model ability.
 - Next time, I will ask someone to solve each task using only its instructions before collecting results.
-
- Recorded actions and failure analysis. Preserve the original results. Correct the instructions in a new test version.

Local speed measurements



17.34 ms

Create session • median



1.50 ms

Reset state • median



84.15 ms

Screenshot • median

- These measurements used an Apple M5 Max with Node 24.
 - The test created 100 sessions and ran 400 reads, with up to 16 reads at once.
 - The browser test used 10 sessions, one after another.
 - These times exclude model calls. They do not measure how many agents can run together.
 - Next, test a queue of runs across separate workers with shared recording storage. Measure each worker's full memory use.
-
- Timing samples and test setup. The multi-worker design is proposed, not measured.

Other mistakes I made

- I did not establish suitable limits for the longest task. Every retrospective run reached a limit.
 - I let optional screenshot failures stop text-based agents. Recording failures now stay separate from task results.
 - I initially displayed provider errors as task failures. The UI now shows them as blocked runs.
 - I missed late replay updates that could replace the current view. Run ownership checks now reject those updates.
 - I did not secure enough disk headroom. The safety check stopped collection before all runs were recorded.
-
- [Build review: evidence, fixes, and remaining work](#). Collection later finished without replacing any recorded attempt.

Demo and questions

- [Try the workspace](#). Search, reply in a thread, and edit a message.
 - [Watch a recorded run](#). Inspect the actions, workspace changes, and result checks.
 - Which workflows should we add next?
 - Should the score penalize wrong actions even when the agent undoes them?
 - How many agents must run at once, and what cost can we allow?
-
- Relay can reset, execute actions, and return a reward. No reinforcement learning training has been run.
 - [Code and setup](#) · [Requirements and remaining work](#).